Improved YAML syntax for automations - PDF Guide

With one of the latest Home Assistant updates, the YAML syntax for automations has been improved and made more intuitive. YAML (Yet Another Markup Language) is the file format Home Assistant uses to store automations. These changes aim to make the code easier to understand and work with. But is the new syntax really simpler, or are we dealing with something more complex? In this guide, we'll take a deep dive into these changes, compare them with the old style, and help you get comfortable with writing and updating automations.

Section 1: Why Change the YAML Syntax?

First things first: don't panic! The new YAML syntax is an improvement, but it **won't break any of your existing automations**. That means you can continue using the old YAML syntax if you prefer it or if your current setup works perfectly fine. Home Assistant has promised that the old syntax will remain compatible, so there's no need to rush and update every single automation.

However, Home Assistant recommends using the new YAML syntax for any new automations, and the new syntax will be automatically used when Graphical Automation Editor is used. Why? Because the new syntax is designed to be more natural, readable & easier to understand. If you're just getting started with automations, or if you want to future-proof your setup, it's a good idea to start learning this new format now.

Key Components of Automations

Now, before we start comparing the new syntax with the old, let's break down the key components of an automation.

- **Triggers** (required): Triggers are what start the automation. For example, they can be based on time, sensor changes, or a person arriving home. You can add multiple triggers, and the automation will start when **any** of these triggers happen.
- **Conditions** (optional): Conditions are rules that need to be true for the automation to run. If you have multiple conditions, by default, **all** of them need to be true. However, you can change this behavior by using logical operators like "or" or "and."
- **Actions** (required): Actions are the tasks that the automation will perform when triggered. You can list multiple actions, and they will run in sequence. For example, the actions might include turning on lights, sending a notification, or adjusting the thermostat.

Section 2: What Has Changed in the YAML Syntax?

Let's get into the changes. The new YAML syntax for automations has undergone a few key updates to make it more intuitive and reflect the way automations work. Here's a summary of the main changes:

1. **Triggers**: The word "trigger" has been changed to **triggers**.
2. **Conditions**: The word "condition" is now **conditions**.
3. **Actions**: The word "action" is now **actions**.

Why these changes? Automations often involve multiple triggers, conditions, or actions, so using the plural form makes more sense and better represents how automations are structured. For instance, instead of having just one trigger, you can have several (such as time-based triggers, motion sensor triggers, or someone arriving home). Similarly, you can have multiple conditions that must all be true before the automation will run, and a series of actions that will be carried out once the automation is triggered.

In the old syntax, these keys (trigger, condition, action) were singular, which made sense for simpler automations but wasn't very clear for automations that had multiple steps. By switching to the plural form, it's now more obvious that these sections can hold a list of items.

Another important change is in the **trigger definition**. Previously, the trigger section had a key called **platform** to specify the type of trigger (such as time, state, or event). In the new syntax, this is simplified to just **trigger**, which makes the code cleaner and easier to read.

Section 3: What Does This Mean for You?

So, how do these changes affect you? The new YAML syntax provides you with the flexibility to define:

- **Multiple triggers**: Your automation can be triggered by different events (for example, time of day, motion detection, or a change in someone's location).
- **Multiple conditions**: You can define several rules that need to be true before the automation runs (such as the time being between 6 PM and 11 PM, or the house being occupied).
- **Multiple actions**: You can create a sequence of actions that will be performed in order (such as turning on lights, adjusting the thermostat, or sending a notification).

Let's now compare an example of the old YAML syntax with the new syntax so you can see the differences in action.

Example of an Automation (New YAML Syntax)

Here's an example of an automation in YAML using the new syntax. This automation will turn on your lights one hour before sunset if someone is home. It will also turn on the lights if someone arrives home between 4 PM and 11 PM.

```
automation my_lights:
  # Turns on lights 1 hour before sunset if people are home
  # and if people get home between 16:00-23:00
  - alias: "Rule 1: Light on in the evening"
    triggers:
      # Prefix the first line of each trigger configuration
      # with a '-' to enter multiple
      - trigger: sun
        event: sunset
        offset: "-01:00:00"  # 1 hour before sunset
      - trigger: state
        entity_id: all
        to: "home"  # when someone arrives home
    conditions:
      # Prefix the first line of each condition configuration
      # with a '-'' to enter multiple
      - condition: state
        entity_id: all
        state: "home"  # only if someone is home
      - condition: time
        after: "16:00:00"  # between 4 PM
        before: "23:00:00"  # and 11 PM
    actions:
      # With a single action entry, we don't need a '-' before
action - though you can add it if you want to
      - action: homeassistant.turn_on
        target:
          entity_id: group.living_room  # turns on the living
room lights
```

How Does This Compare to the Old Syntax?

Now, let's look at how this automation would have been written in the old YAML syntax, before the recent update.

```
automation:
  - alias: "Rule 1: Light on in the evening"
    trigger:
```

```
    - platform: sun
      event: sunset
      offset: "-01:00:00"
    - platform: state
      entity_id: all
      to: "home"
  condition:
    - condition: state
      entity_id: all
      state: "home"
    - condition: time
      after: "16:00:00"
      before: "23:00:00"
  action:
    - action: homeassistant.turn_on
      entity_id: group.living_room
```

Let's break down the differences:

1. **Triggers**:
   - In the **old syntax**, we used the singular key `trigger`. You can also see that we had to specify `platform: sun` and `platform: state` to define the type of trigger.
   - In the **new syntax**, `trigger` has become **triggers** to represent that multiple triggers can be used. Also, instead of using the word `platform`, we directly start with `trigger: sun` and `trigger: state`, making the code easier to read and less cluttered.
2. **Conditions**:
   - Similarly, the old syntax uses a singular `condition` key.
   - The new syntax uses **conditions**, making it clear that you can list several conditions.
3. **Actions**:
   - In the super old syntax (before Home Assistant 2024.8), we had the `action` key and used `service: homeassistant.turn_on` to define what action to take. After 2024.8 `service:` was replaced with `action:`
   - In the new and latest syntax so far `action` is replaced with **actions.**

Additional Information: Editing YAML Automations in Home Assistant

Even though Home Assistant provides a graphical interface to create automations, all of them are saved in YAML format behind the scenes. You can easily edit this YAML code if you want more control or need to fine-tune your automations. Here's how:

1. Open Home Assistant.
2. Go to **Settings** > **Automations & Scenes**.
3. Select the automation you want to edit.
4. Click the menu (three dots) in the top-right corner.
5. Choose **Edit in YAML**.

This allows you to see the YAML code and make direct changes.

Conclusion

The new YAML syntax in Home Assistant is designed to make automations clearer, more intuitive, and easier to work with. By using plural keys like **triggers**, **conditions**, and **actions**, it's now more obvious that you can list multiple items in each section. This new format also reduces the amount of repetitive code, making automations more readable.

While the old syntax will continue to work, learning the new format is a good idea, especially for future automations. With a little practice, you'll be able to write, edit, and manage even the most complex automations with ease. Don't forget to expect more e-mails from me with more useful Smart Home & Home Assistant content like this one!

Home Assistant YAML Automation Quiz

Question 1:
What is the main reason for Home Assistant updating the YAML syntax for automations?
A) To reduce the file size
B) To make the code more intuitive and easier to read
C) To make it faster to process
D) To remove old features

Question 2:
Which of the following is correct in the new YAML syntax?
A) `run`
B) `platform`
C) `trigger`
D) `service`

Question 3:
True or False: The old YAML syntax for automations will stop working after the update.

Question 4:
What is the new term used to define multiple steps that trigger an automation?
A) `action`
B) `platforms`
C) `conditions`
D) `triggers`

Question 5:
In the new YAML syntax, how do you define multiple conditions that need to be true for an automation to run?
A) By using the key `conditions`
B) By using the key `condition`
C) By using the key `triggers`
D) By using the key `action`

Question 6:
Which part of the following automation is incorrect based on the new YAML syntax?

```
automation my_example:
  - alias: "Example Automation"
    trigger:
      - trigger: state
        entity_id: all
        to: "home"
    condition:
      - condition: time
        after: "08:00:00"
        before: "22:00:00"
    actions:
      - action: homeassistant.turn_on
        target:
          entity_id: group.kitchen_lights
```

A) The key `trigger` should be `triggers`
B) The key `condition` should be `conditions`
C) The key `actions` is missing the dash (-)
D) Both A and B

Question 7:
Which key is used to list the tasks that will happen after the automation is triggered?
A) `variables`
B) `id`
C) `actions`
D) `condition`

Question 8:
How do you specify when an automation should be triggered based on the sunset event in the new YAML syntax?
A) Using `trigger: sun`
B) Using `trigger: time`
C) Using `platform: sun`
D) Using `trigger: event`

Question 9:
Which section of YAML can contain optional conditions to check before performing actions?
A) `triggers`
B) `conditions`
C) `actions`
D) `id`

Answers

Answer 1:
B) To make the code more intuitive and easier to read

Answer 2:
C) `trigger`

Answer 3:
False – The old YAML syntax will continue to work.

Answer 4:
D) `triggers`

Answer 5:
A) By using the key `conditions`

Answer 6:
D) Both A and B – The new syntax uses `triggers` and `conditions`, not `trigger` and `condition`.

Answer 7:
C) `actions`

Answer 8:
A) Using `trigger: sun`

Answer 9:
B) `conditions`

How Did You Do?

- **8-9 correct**: YAML master! You've got a great grasp of the new syntax and Home Assistant automations.
- **6-7 correct**: Well done! You're on your way to mastering YAML automations.
- **4-5 correct**: Good effort! A little more practice and you'll be there.
- **3 or below**: Don't worry—review the material and try the quiz again!